Rafeed Mohammad Sultan

AI Researcher & Software Engineer

+880-1732073478 sultanrafeed@gmail.com Portfolio GitHub Rafeed Sultan

Dhaka, Bangladesh

Research Interests

I am passionate about advancing Artificial Intelligence, focusing on Natural Language Processing (NLP). My research interests include Trustworthy/Responsible AI, focusing on LLM alignment and evaluation, prompt optimization for robust instruction following and tool use, and retrieval-augmented generation (RAG) with emphasis on factuality, safety, and multilingual/low-resource settings. I also pursue Medical AI applications of NLP for clinical text and decision support, with the broader goal of developing transparent, fair, and accountable language technologies aligned with human values.

Research Publications

- **Beyond Labels: Aligning Large Language Models with Human-like Reasoning**
  International Conference on Pattern Recognition (ICPR) 2024
  GitHub arXiv **[Accepted]**

- **Efficient Skin Cancer Detection Using Lightweight Model Souping and Ensembling Knowledge Distillation for Memory-Constrained Devices**
  ScienceDirect
  Q1 Journal: Intelligence-Based Medicine **[Accepted]**

- **Empowering Meta-Analysis: Leveraging Large Language Models for Scientific Synthesis**
  2024 IEEE International Conference on Big Data
  GitHub arXiv **[Accepted]**

- **LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval**
  IJCNN 2025: International Joint Conference on Neural Networks
  arXiv **[Accepted]**

- **Paper**
  Pattern Recognition Letters **[Under Process]**

Education

- **BSc in Computer Science Engineering** *2020-2024*
  North South University, Dhaka, Bangladesh CGPA: 3.38/4.00

- **Edexcel A Levels** *2019*
  Cephalon International School, Dhaka, Bangladesh
  Subjects: Physics, Chemistry, Biology, Mathematics

- **Edexcel O Levels** *2017*
  Academia, Dhaka, Bangladesh
  Subjects: English Language, Bangla, Mathematics B, Pure Mathematics, Chemistry, Biology, Human Biology

Research Experience

Cross-Model Evaluation: Judging AI Ethics and Alignment Responses with Language Models
Addressed ethical concerns in LLMs using the Dataset for Aligning Reasons (DFAR), which enhances the ETHICS dataset with detailed annotations. Utilized LLMs to assess and reason ethicality in DFAR statements, automating evaluations using other LLMs. Findings showed strong alignment between LLM and human evaluations, demonstrating reliability and scalability. [GitHub]

Optimizer-Performance-Evaluation
Investigated the impact of different optimizers and batch sizes on the performance of a custom CNN built with VGG blocks. [GitHub]

Predictive Modeling and Explainable AI Analysis of Farm Features and Practices Impacting White Spot Disease Prevalence in Farmed Shrimps in Bangladesh
Developed a machine learning model with 72.67% accuracy to predict White Spot Disease prevalence in shrimp farms and used SHAP to analyze the impact of farming practices on disease trends. [GitHub]

Efficiency Comparison of Programming Languages in Algorithm Development
Aimed to determine which programming language is most efficient in terms of runtime efficiency, memory consumption, and CPU utilization by implementing various algorithms with differing time complexities in different programming languages. [GitHub]

Experience

•Software Engineer *Apr 2026 – Present*
SELISE Digital Platforms Dhaka, Bangladesh
- Leading development of **OpenClaw** and expanding **MCP** server tooling; driving architecture decisions across agentic AI and enterprise automation workstreams.

•Associate Software Engineer *Jan 2025 – Mar 2026*
SELISE Digital Platforms Dhaka, Bangladesh
- Built a voice agent: **ElevenLabs** TTS, rigorous prompt engineering (guardrails, dynamic variables, refusals), **MCP** tool server, and **NestJS** + **MongoDB** services; automated GitHub issue creation via signed webhooks.
- Led agentic chat platform: real-time WebSocket streaming, Markdown rendering, and a React–Flask ingestion pipeline for PDF/image context into Azure PostgreSQL (preview, enable/disable, delete); explored SAML-based SSO.
- Developed energy-trading analytics (**PCAT**): DAM/RTM PnL and PTP/AS exposure; implemented calculations with pandas/SQLAlchemy and PostgreSQL schemas/ETL.
- Authored technical docs and evaluation protocols; shipped automated NPS workflows to reduce manual operations.

•Intern Software Developer *Oct 2024 – Jan 2025*
SELISE Digital Platforms Dhaka, Bangladesh
- Supported finance/procurement automation and ops tooling to improve cycle time and data quality.

•Part-time Content Writer *Sep 2023 – Nov 2023*
GameRiv Online
- Produced SEO-aware articles on deadline; maintained publication quality in WordPress.

•Graphics Designer *Jan 2020 – Jun 2020*
Signature Minds Institute Online
- Delivered client assets on schedule; collaborated on visual branding using Canva.

•Teacher's Assistant *Nov 2019 – Jun 2020*
Study Town Lalmatia, Bangladesh
- Graded A-level Biology/Chemistry and tutored students in core topics.

Projects

Rokkha (Android App) - An app to enhance safety by alerting guardians with the user's location at the press of a button. Users can set intervals for periodic location alerts to specified contacts. *Technologies: Kotlin, Firebase, SQLite* [GitHub]

Roomafy (Web App) - A collaborative music-playing app integrated with Spotify API. *Technologies: Django, React, HTML+CSS, Bootstrap* Ongoing

Phonebook (Web App) - A mini project in Java for storing and managing contacts. *Technologies: Java Swing, SQLite* [GitHub]

Library Management System (Web App) - An online platform for storing PDF books and managing library resources. *Technologies: HTML+CSS, MySQL, PHP, Bootstrap* [GitHub]

Automation With Selenium - Automated a website for GigaTech, generating test cases and ensuring efficient testing. *Technology: Python (Selenium)* [GitHub]

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++
ML / AI: PyTorch, Hugging Face Transformers, PEFT, Datasets, scikit-learn, FAISS, pandas, NumPy
NLP: LLM fine-tuning, RAG (dense + sparse retrieval), prompt engineering, LLM alignment & evaluation
Back-end: NestJS, Flask, Django, PostgreSQL, MongoDB, SQLAlchemy, psycopg2, REST APIs
Front-end: React, HTML, CSS, WebSocket
Cloud & DevOps: Azure (PostgreSQL, Blob), Firebase, Git, GitHub, Postman
Agentic / LLM Tooling: Model Context Protocol (MCP), ElevenLabs, LangChain, OpenAI API
Other: LaTeX, Overleaf, WordPress

Awards & Achievements

Daily Star Award for outstanding O levels results.

Positions of Responsibility

•**Volunteer** Jaago Foundation - Dhanmondi, Dhaka *May 2016*

– Raised Tk. 20,000+ for youth education and well-being through community outreach and fundraising efforts.

Certifications & Workshops

IBM via Coursera *(2024)* — Tools for Data Science, Python for Data Science AI & Development, Data Science Methodology, Databases & SQL for Data Science (Honors), Python Project for Data Science, What is Data Science?, Introduction to Software Engineering

Google via Coursera *(2024)* — Prepare Data for Exploration, Ask Questions to Make Data-Driven Decisions, Foundations: Data, Data, Everywhere

Workshops — Quantum Mechanics & Quantum Computing by Dr. Mahdy Rahman Chowdhury *(May 2022)*, Deep Learning with PyTorch by Hasib Zunair *(Jan 2021)*

Duke University via Coursera — Programming Fundamentals *(Jan 2020)*

References

Nabeel Mohammed, PhD
Associate Professor
Dept. of Electrical & Computer Engineering
North South University
nabeel.mohammed@northsouth.edu

Arifur Rahman
Principal Software Architect
SELISE Digital Platforms
arifur.rahman@selisegroup.com

Shahriar Azad
Financial Technology Consultant
SELISE Digital Platforms
shahriar.azad@selisegroup.com

Anshah Anju Khan Chowdhury
VP of Finance & Legal
SELISE Digital Platforms
anshah.chowdhury@selisegroup.com